

Private Markets Analytics Pipeline

SQL Server 2022 · dbt core · Power BI

Project Supplement · **Analytics Engineering**

Independent Portfolio Project · 2026

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About This Document

What this is

An independent portfolio project demonstrating modern analytics engineering practices applied to a private markets reporting context.

The project is a complete end-to-end private markets analytics pipeline built from scratch in 2026 to demonstrate applied analytics engineering in a PE fund data context.

All code, models, and documentation were written independently using realistic private markets data structures and finance logic.

What it covers

- 10 source tables loaded from CSV via rerunnable BULK INSERT scripts
- End-to-end 8-layer pipeline: raw landing → dbt staging → intermediate → star schema → mart
- dbt intermediate models: J-curve backbone, IRR input shaping, quarterly snapshots
- Star schema: dim_date, dim_fund, dim_investor + fact_cashflows, fact_nav, fact_commitments
- Reporting mart: DPI, RVPI, TVPI, break-even flag — all pre-computed in CHF
- Post-model validation: TVPI = DPI + RVPI identity check + 4 other SQL validation scripts
- Power BI: Portfolio Overview page + J-Curve & Cashflow page

Stack Alignment — Target Stack vs. This Project

Target Stack

Modern data engineering roles

Snowflake Cloud data warehouse (primary)

dbt Transformation + modelling layer

Power BI Reporting and visualisation

SQL Transformation and query logic

Private Markets PE / infrastructure fund context



This Project

Private Markets Analytics Pipeline

SQL Server 2022 Local dev — dbt is warehouse-agnostic

dbt core 100% identical — models, YAML, SQL

Power BI 100% identical stack

SQL Complex finance SQL throughout

Private Markets Fund economics, NAV, IRR, J-curve

Note: dbt is warehouse-agnostic. Switching SQL Server → Snowflake requires only a profile adapter change. All models, YAML, and SQL remain identical.

Business Problem — Why PE Platforms Are Not Analytics-Ready

Private equity teams run operations on platforms like eFront, Allvue, and iLEVEL — workflow and accounting tools, not analytics engines.

No Star Schema

Data lives in operational tables built for data entry and workflow. There is no dimensional model, no clean join structure, and no portfolio-level aggregation layer.

No Portfolio Aggregation

Computing fund-level TVPI, DPI, RVPI, or IRR requires manual Excel extraction. No reliable single source of truth across funds, vintages, or strategies.

No SQL-Accessible Layer

Reporting is locked inside the platform UI. Teams cannot write custom SQL, build downstream dashboards, or version-control their transformation logic.

Solution: build a proper analytics layer on top — dbt models the data into a star schema, Power BI consumes it. No platform dependency.

Solution Architecture — End-to-End Pipeline (8 Layers)

1

Source Data

10 CSV exports: funds, investors, commitments, cashflows, NAV, assets, FX rates, date spine

2

Raw Layer

SQL Server: rerunnable BULK INSERT scripts, 10 tables, ~34,000 rows total across all sources

3

Staging (dbt)

stg_ models: type casting, renaming, NULL handling, date parsing — grain preserved

4

Intermediate (dbt)

int_ models: J-curve cumulative series, IRR input shaping, quarterly snapshot rollups

5

Star Schema (dbt)

dim_date, dim_fund, dim_investor + fact_cashflows, fact_nav, fact_commitments

6

Reporting Mart (dbt)

mart_fund_quarterly_performance: DPI, RVPI, TVPI, break-even flag — 1 row = 1 fund × quarter

7

Validation Layer

Post-build SQL checks: J-curve integrity, IRR input structure, TVPI = DPI + RVPI identity

8

Power BI

Portfolio Overview (KPIs, NAV, TVPI) + J-Curve & Cashflow (paid-in, distributions, J-curve)

Raw Data Layer — 10 Source Tables, ~34,000 Rows

All 10 tables are loaded from CSV via rerunnable BULK INSERT scripts (TRUNCATE + BULK INSERT pattern). Loaded to the raw schema in SQL Server 2022.

Transaction & Fact Tables

cashflows_v7

927

Fund-level capital calls and distributions. Source for DPI, RVPI, TVPI, IRR, J-curve.

investor_cashflows_v7

10,004

LP-level cashflow allocations with commitment share %, allocation method.

nav_v7

156

Quarterly NAV snapshots per fund. Core stock measure for RVPI and break-even logic.

commitments_v7

55

LP commitment events per fund with MFN, side-letter, and fee terms flags.

asset_quarterly_snapshot_v7

765

Asset-level quarterly data: CumulativeInvested, GrossMOIC, FairValue, RealizationStatus.

Master & Reference Tables

funds_v7

5

Fund master: strategy, economics (mgmt fee, carry, hurdle), ESG article, lifecycle dates.

investors_v7

11

LP master: investor type, region, country, ESG preference, tax status.

asset_master_v7

30

Portfolio company master: AssetType, Sector, GrossMOICTarget2025, InvestmentType.

date_dim_v7

3,653

Conformed calendar date spine: year, quarter, month, flags (quarter-end, year-end).

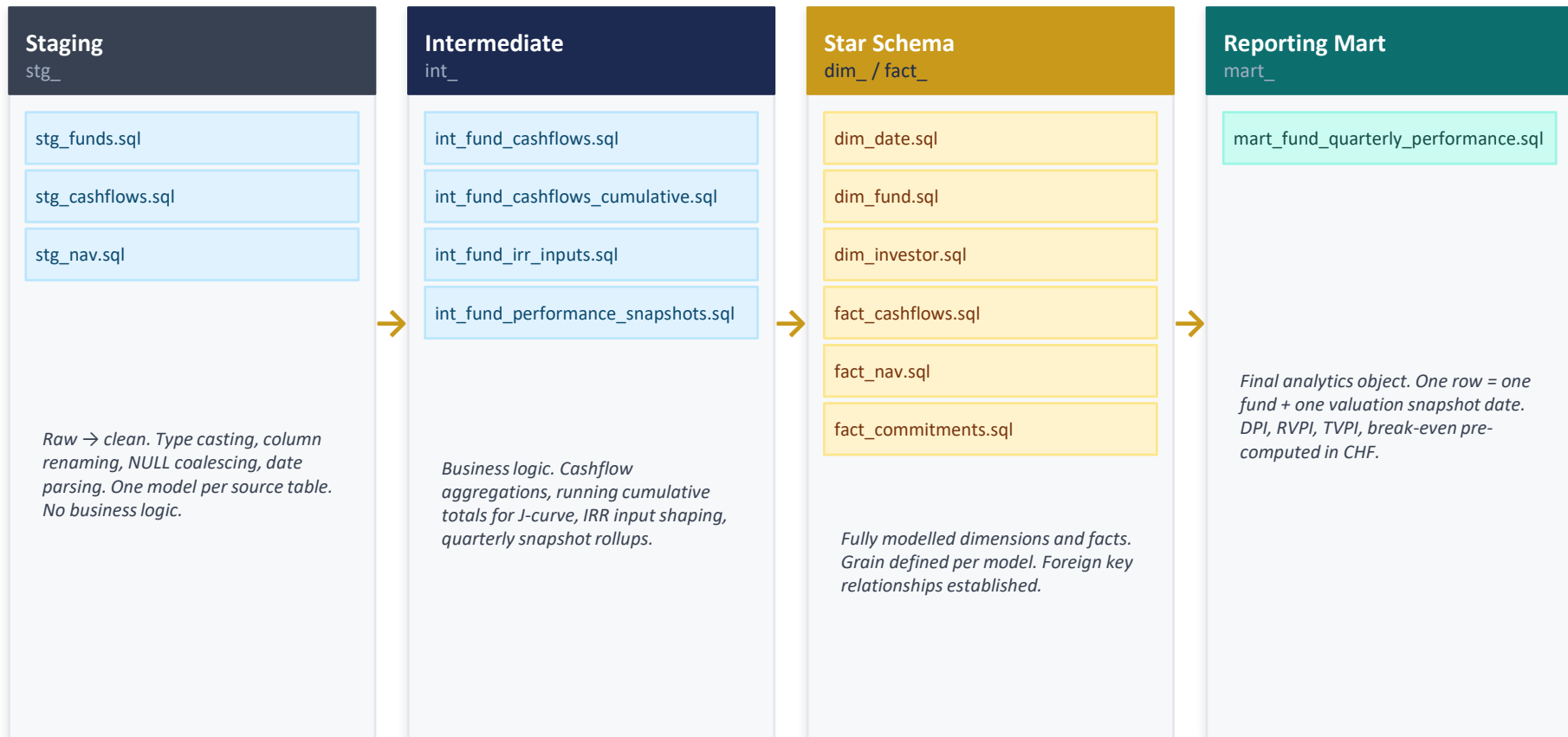
fx_rates_chf_v7

18,265

Daily FX rates to CHF for all fund currencies. Applied at transaction and NAV level.

Total rows loaded: ~34,000 · Largest: fx_rates_chf_v7 (18,265 daily FX rates) · Date spine: 3,653 rows · All 10 tables loaded via rerunnable BULK INSERT (TRUNCATE + INSERT)

dbt Model Architecture — Staging → Intermediate → Star Schema → Mart



Star Schema Design — 3 Dimensions + 3 Facts

dim_date

- date_id (PK)
- full_date
- year_quarter
- is_quarter_end

dim_fund

- fund_id (PK)
- fund_name
- strategy
- vintage_year

dim_investor

- investor_id (PK)
- investor_name
- investor_type
- domicile

fact_cashflows

Grain: fund + investor + date + type

- fund_id (FK)
- investor_id (FK)
- date_id (FK)
- amount_chf

fact_nav

Grain: fund + valuation date

- fund_id (FK)
- date_id (FK)
- nav_amount_chf
- fx_rate_to_chf

fact_commitments

Grain: fund + investor

- fund_id (FK)
- investor_id (FK)
- commitment_amount_chf
- commitment_pct_of_fund

FK joins

SQL Transformation Logic — Finance Metrics

J-Curve: Cumulative Net Cashflow

```
-- int_fund_cashflows_cumulative
SUM(net_cashflow_amount)
OVER (
  PARTITION BY fund_id
  ORDER BY cashflow_date
  ROWS BETWEEN UNBOUNDED
  PRECEDING AND CURRENT ROW
)
AS cumulative_net_cashflow
```

Running cumulative net cashflow per fund over time. Calls are negative; distributions positive. The J-curve shape emerges from this series.

Cumulative Paid-In & Distributions

```
CASE WHEN amount < 0
  THEN -amount ELSE 0
END AS paid_in_amount,

CASE WHEN amount > 0
  THEN amount ELSE 0
END AS distribution_amount,

-- Rolled up at snapshot grain:
COALESCE(SUM(paid_in_amount_chf), 0)
AS cumulative_paid_in_chf
```

Calls recast as positive paid-in; distributions kept positive. Rolled up to quarter-end snapshot grain for DPI/RVPI/TVPI calculation.

DPI / RVPI / TVPI

```
-- mart_fund_quarterly_performance
CAST(cumul_distributions_chf
  AS decimal(38,10))
/ NULLIF(CAST(cumul_paid_in_chf
  AS decimal(38,10)), 0)
AS dpi,

nav_amount_chf
/ NULLIF(cumul_paid_in_chf, 0)
AS rvpi,

(nav_amount_chf + cumul_distributions_chf)
/ NULLIF(cumul_paid_in_chf, 0)
AS tvpi
```

Core PE return multiples in CHF. TVPI = DPI + RVPI identity validated post-build. NULLIF guards against division-by-zero before first call.

IRR: shaped in int_fund_irr_inputs — actual cashflows (ACTUAL_CASHFLOW) + terminal NAV row (TERMINAL_NAV) per valuation snapshot, ready for XIRR-equivalent calculation.

Reporting Mart — mart_fund_quarterly_performance

Grain: one row = one fund + one valuation snapshot date · All metrics computed in CHF reporting currency

Fund & Period	Capital Activity	Valuation	Performance KPIs
fund_id	cumulative_paid_in_chf	nav_amount_chf	dpi
fund_name	cumulative_distributions_chf	unfunded_commitment_chf	rvpi
strategy	cumulative_net_cashflow_chf	total_value_chf	tvpi
vintage_year	paid_in_pct_of_fund_size_chf	value_creation_chf	break_even_flag
valuation_date		nav_pct_of_fund_size_chf	valuation_status
quarter_label			

Built on top of fact_nav + fact_cashflows + dim_fund + dim_date. All KPIs pre-computed in SQL — Power BI consumes this as a thin semantic layer with minimal DAX.

Business Questions Answered — From Mart to Insight

The reporting mart answers fund manager questions directly with a single SELECT. No Excel, no platform export, no manual calculation.

Which funds have returned their capital to LPs?

Flag is set to 1 when $\text{total_value (NAV + distributions)} \geq \text{cumulative paid-in}$. Evaluated at each quarter-end snapshot.



break_even_flag = 1

mart_fund_quarterly_performance

What is each fund's return multiple — and how does TVPI decompose?

$\text{DPI} = \text{distributions} / \text{paid-in}$ · $\text{RVPI} = \text{NAV} / \text{paid-in}$ · $\text{TVPI} = \text{DPI} + \text{RVPI}$. Identity validated post-build. All in CHF.



dpi, rvpi, tvpi

mart_fund_quarterly_performance

How much of committed capital has been called to date?

$\text{cumulative_paid_in_chf} \div \text{fund_size_chf_final_close}$. Shows deployment pace and remaining unfunded exposure.



paid_in_pct_of_fund_size_chf

mart_fund_quarterly_performance

How does the J-curve evolve for each fund over time?

Running sum of net cashflows by fund and date via `SUM() OVER (PARTITION BY fund_id ORDER BY cashflow_date ROWS UNBOUNDED PRECEDING)`.



cumulative_net_cashflow

int_fund_cashflows_cumulative

What is the total portfolio NAV trend by quarter?

Sum of `nav_amount_chf` across all funds grouped by `year_quarter`. Rendered as stacked area or line chart in Power BI Portfolio Overview.



nav_amount_chf · year_quarter

mart_fund_quarterly_performance

All five questions are answerable with a single SELECT against `mart_fund_quarterly_performance` — no joins, no subqueries, no DAX. The mart is the analytics contract between SQL and Power BI.

Mart Output — mart_fund_quarterly_performance (Sample Rows)

1 row = 1 fund × 1 quarter-end valuation date · 156 snapshot rows across 5 funds · Directly consumed by Power BI as a SQL Server view

fund_name	val. date	nav_chf	paid_in_chf	dpi	rvpi	tvpi	break_even	strategy
Alpha Buyout I	2024-09-30	CHF 42.0M	CHF 150.0M	1.062	0.280	1.342	✓ YES	Buyout
Beta Growth II	2024-09-30	CHF 80.4M	CHF 80.0M	0.537	1.005	1.542	✓ YES	Growth
Gamma Infra I	2024-09-30	CHF 189.8M	CHF 200.0M	0.232	0.949	1.181	✓ YES	Infrastructure
Delta Direct Lending I	2024-09-30	CHF 22.2M	CHF 70.0M	0.740	0.317	1.057	✓ YES	Direct Lending
Epsilon Real Estate I	2024-09-30	CHF 70.7M	CHF 120.0M	0.526	0.589	1.114	✓ YES	Real Estate

Latest snapshot per fund (2024-09-30). DPI, RVPI, TVPI from live mart_fund_quarterly_performance view; TVPI = DPI + RVPI identity validated post-build. CHF values computed via fx_rates_chf_v7 join at each cashflow and NAV date. break_even_flag: 1 when (nav + distributions) ≥ cumulative paid-in.

Validation Framework — Post-Build SQL Checks

A structured set of post-build validation scripts confirms the pipeline is structurally correct and financially coherent at every layer.

Raw Load Checks

4 checks

- ✓ Row counts match source CSV expected values
- ✓ All 10 tables present in raw schema
- ✓ First 10 rows preview of funds_v7, cashflows_v7, investor_cashflows_v7
- ✓ Schema: exactly 10 user tables in raw schema

Intermediate Checks

4 checks

- ✓ J-curve backbone: cumulative_net_cashflow starts negative for PE-style funds
- ✓ Cumulative path evolves correctly in chronological order per fund
- ✓ IRR inputs: ACTUAL_CASHFLOW rows present before TERMINAL_NAV per snapshot
- ✓ IRR input shape: exactly one TERMINAL_NAV row per fund x valuation date

Mart Structure

3 checks

- ✓ mart_fund_quarterly_performance exists as SQL Server view in mart schema
- ✓ Latest snapshot per fund: 1 row per fund at most recent valuation date
- ✓ IRR input rows: actual cashflows + terminal NAV both present per snapshot

Financial Logic

4 checks

- ✓ First 50 rows review: dpi/rvpi/tvpi/break_even_flag look sensible
- ✓ Latest snapshot per fund: cumulative_paid_in, nav, total_value reviewed
- ✓ TVPI = DPI + RVPI identity: $ABS(tvpi - (dpi+rvpi)) < 0.000001$ for all rows
- ✓ Full mart preview: 1,000 rows ordered by fund_id, valuation_date

Power BI Report — Portfolio Overview + J-Curve & Cashflow

Two report pages consuming mart_fund_quarterly_performance + int_fund_irr_inputs. Thin semantic layer — all logic lives in SQL, not DAX.

Page 1 — Portfolio Overview

KPI Card Row

Total NAV (CHF) · TVPI · DPI · RVPI · Fund count · Total commitments

TVPI by Fund

Horizontal bar chart — compare return multiples across all funds side by side

Portfolio NAV Composition

Stacked area chart — NAV contribution per fund by quarter over time

Portfolio NAV Trend

Line chart — total portfolio NAV evolution across all quarter-end snapshots

Page 2 — J-Curve & Cashflow Analysis

Paid-In Bars

Bar chart — quarterly capital calls (outflows) per fund. Shows deployment pace over time.

Distribution Bars

Bar chart — quarterly distributions (inflows) per fund. Shows return activity by period.

Cumulative Net Cashflow

Line chart — the J-curve path. Starts negative (capital calls), crosses zero when fund breaks even.

Fund Slicer

Single-fund drill-in — inspect the full cashflow and valuation history for any one fund.



Microsoft Power BI





Microsoft Power BI



Annex — SQL File Index

Raw Landing (SQL Server) (4 files)

01_platform_setup__create_database_and_all_core_schemas.sql

02_raw_landing__create_raw_tables.sql

03_raw_landing__load_raw_tables_from_csv.sql

04_raw_landing__validate_raw_load.sql

Staging (stg_) (3 files)

stg_funds.sql

stg_cashflows.sql

stg_nav.sql

Intermediate (int_) (4 files)

int_fund_cashflows.sql

int_fund_cashflows_cumulative.sql

int_fund_irr_inputs.sql

int_fund_performance_snapshots.sql

Dimensions (dim_) (3 files)

dim_date.sql

dim_fund.sql

dim_investor.sql

Facts (fact_) (3 files)

fact_cashflows.sql

fact_nav.sql

fact_commitments.sql

Reporting Mart (1 file)

mart_fund_quarterly_performance.sql

Post-Build Validation (4 files)

01_validate_int_fund_cashflows_cumulative.sql

03_check_latest_snapshot_per_fund.sql

04_inspect_irr_input_rows.sql

05_validate_rebuilt_mart.sql

Let's connect

The pipeline, dbt models, SQL transformation files, validation scripts, and Power BI reports are available for review at git hub link : https://github.com/gny-analytics/Private_Markets_Analytics_Pipeline

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